

Acknowledgements

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Future Work

Moving forward, it is crucial to refine LLM development and evaluation techniques to better handle evolving informational environments, ensuring their continued relevance and utility in various applications.

Looking ahead, we suggest future work includes several ambitious goals aimed at enhancing the versatility and depth of our evaluation framework. Key among these objectives are:

- Exploring mitigation strategies, through we conducted preliminary investigations into prompt engineering strategies that could potentially mitigate temporal performance degradation, see Abstract B.
- Integrating a wider array of models, particularly focusing on those based on the Transformer architecture, such as Griffin and Mamba models.
- Expanding the number of checkpoints accessed for a more comprehensive temporal analysis of model performance.
- Delving into the relationship between token-level loss metrics and overall scores, which could offer more granular insights into model capabilities.
- Developing methodologies to assess the obsolescence of benchmarks and models, potentially offering a dynamic approach to evaluating the relevance and efficiency of language models over time.

These initiatives represent our commitment to advancing the state of the art in language model evaluation, ensuring that our tools remain at the forefront of technological progress and innovation.

Limitations

While we strive to collect textual data from a variety of sources such as news articles, scientific papers, and forum discussions, the proportion and diversity of the texts may still fall short of fully reflecting the diversity and complexity of language. Also, the English-centric nature of the questions and answers limits the benchmark’s applicability to non-English speaking regions and global events. Moreover, such evaluations on text loss require access to the model’s logits, which is challenging for closed-source models or models accessed via APIs.

Potential Risks

Bias and Fairness Risks: When used to make predictions about future events, language models can inadvertently perpetuate or amplify biases present in their training data. This could lead to biased forecasts that reflect and potentially exacerbate existing societal prejudices. Such biased outputs may result in models that are unfair or discriminatory, which could harm individuals or groups and damage the reputation of the organizations involved.

Ethics Statement

We fully acknowledge the need to adhere to ethical standards in handling and generating textual data. Hence, we commit to:

- We utilize various datasets in our experiments, including publicly available sources like news articles and academic papers, which do not contain personally identifiable information or offensive content. However, some datasets created or used during our study contain sensitive information or are bound by specific usage agreements that prevent public sharing. Therefore, these datasets are utilized strictly within our experiments and are not released as part of our open-source materials, adhering to ethical guidelines and agreements.
- **Responsibility for Content Generated by Open-Source Models:** We recognize that content generated by our system or through the use of open-source models involved in the evaluation might reflect or amplify biases. While efforts are made to minimize these biases and improve the fairness of the content,

we cannot take responsibility for the generated content. We encourage users and developers to critically engage with these models and remain vigilant of their outputs.

- **Addressing Data Leak Issues:** Our system aims to tackle data leak issues by dynamically generating benchmarks and ensuring models generalize to new or future texts. We commit to continuously monitoring and updating our evaluation methods to avoid potential data leakage, ensuring the validity and reliability of our assessment results.
- **Transparency:** We are committed to maintaining the highest level of transparency in our work, including our evaluation methods, data sources used, and assessment outcomes. We encourage open critique and suggestions for improvement of our methodology, aiming to foster knowledge sharing and technological advancement.

We believe that through this approach, we can offer a valuable resource to the research and development community, while fostering deep reflection and discussion on ethical issues in the artificial intelligence field.

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Other text will not be released.

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